

Customer Segmentation Using K-Means and GMM on E-Commerce Transaction Data

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ABSTRACT

The growth of e-commerce transactions has generated large and complex volumes of data, opening up opportunities for companies to gain a deeper understanding of customer behavior patterns. This study aims to segment customers using an unsupervised learning approach based on Recency, Frequency, and Monetary (RFM) features in an online transaction dataset of approximately one million entries, with 700,000 data points meeting the cleanliness criteria. The preprocessing process involves log-transformation and standardization to ensure scale equivalence between features. Using two algorithms, namely K-Means and Gaussian Mixture Model (GMM), the results show differences in accuracy, stability, and segmentation quality. The number of clusters was determined through evaluation metrics such as Silhouette Score for K-Means and Bayesian Information Criterion (BIC) for GMM. The results showed that both algorithms were able to identify customer groups with different shopping behavior characteristics, but GMM provided a more flexible representation of data with high distribution variation. These findings provide a basis for companies to develop more targeted marketing strategies, such as personalized promotions, retention priorities, and customer value-based segmentation.

INTRODUCTION

The development of digital technology and the growth of digital transactions in e-commerce have resulted in a significant volume of data (Nalla and Reddy, 2021). Based on the existing research, around 76% of customers expect personalization in their interactions with brands, but many companies still fail to meet these expectations (Zendesk, 2025). Additionally there is the issue of 50% ineffective digital marketing budgets, due to poor execution, incorrect targeting, and suboptimal use of marketing channels (Proxima, 2025). Moreover, the reality on the ground shows that around 82% customers are willing to share the data needed to get the right personalization (DemandSage, 2025). From cases like this, further analysis can be carried out to determine the needs of the company's strategy in each customer category.

In the context of e-commerce, the use of Big Data and Business Intelligence-based data analysis is becoming increasingly important to support data-driven decision making (John et al., 2023). One of the most widely used analytical approaches to understanding customer behavior is the concept of customer segmentation (Abbasimehr and Shabani, 2021). Customer segmentation aims to group customers based on categories that have similar characteristics, such as purchasing behavior, transaction frequency, or economic value provided to the company (Yan and Zhao, 2021). Through proper segmentation, companies are expected to be able to develop personalized marketing strategies and also increase customer satisfaction.

In general, customer segmentation is carried out using a demographic approach or surveys and questionnaires. However, these methods have limitations, especially in capturing the rapidly changing dynamics of customer behavior in the digital market (Yan and Zhao, 2021). Therefore, an approach based on actual transaction data is a more representative alternative (Miraftabzadeh et al., 2023). With the availability of large amounts of transaction data and various variables, such as purchase value (monetary), transaction frequency, and time of last purchase (recency), machine learning-based analysis methods such

as clustering are highly relevant for producing more accurate customer segmentation (John et al., 2023).

Based on the several previous studies, researchers have explored customer segmentation using the K-Means and RFM (Recency, Frequency, Monetary) approaches. The first study about “Customer Segmentation Using K-Means Clustering: Analyzing RFM Metrics for Marketing Strategies” used a dataset containing 62,667 with transaction periods ranging from April 2021 to August 2023 and applied the K-Means algorithm combined with RFM analysis (Yunita and Ali, 2025). This research was conducted on frozen food retail companies with problems such as a decline in sales without a clear cause, low customer retention and loyalty, and a lack of understanding of customer behavior. The research identified three main customer groups, namely General Development, General Maintenance, and Important Maintenance (Yunita and Ali, 2025).

The second research about the study of “Visualization of Customer Segmentation Based on RFM Attributes Using the K-Means Algorithm” utilized a larger dataset of 541,909 entries and implemented the K-Means method. The primary focus of this research was to visualize the segmentation results through an interactive dashboard using Looker Studio, providing a clearer and more intuitive understanding of customer data (Isnain et al., 2025). The features used are Recency (days since last purchase), Frequency (number of transactions), and Monetary (total transaction value). Two clustering methods are used, namely K-means and K-medoids, with three cluster determination methods, namely silhouette score, Calinski–Harabasz Index, and Davies Bouldin Score (Isnain et al., 2025).

Finally, the research about “Customer Segmentation Using Machine Learning Model: An Application of RFM Analysis” analyzed a dataset within 36,507 records using a combination of K-Means and RFM methods (Pratiwi et al., 2025). The study successfully divided customers into four main segments Platinum, Gold, Silver, and Bronze which serve as a foundation for more personalized and effective marketing strategies (Pratiwi et al., 2025). This journal focuses on the K-means and DBSCAN algorithm methods. Through this research, it is hoped that outlier problems can be overcome by using Box-Cox Transformation.

Based on the review of previous studies, several research gaps can be identified. First, existing customer segmentation studies largely lack explainability mechanisms, limiting the transparency of clustering results for business decision-making. Second, many reported clusters exhibit limited cohesiveness and distinctiveness in terms of customer behavior, reducing their practical value despite acceptable validation metrics. Third, prior research predominantly relies on basic RFM features, with minimal exploration of RFM-derived attributes that could better capture complex customer dynamics. Finally, the absence of business-oriented visualization approaches further hinders the interpretability of customer segments and their translation into actionable marketing strategies. These gaps highlight the need for a more interpretable, robust, and visually intuitive customer segmentation framework.

To address these gaps, this study proposes an integrated customer segmentation framework that combines RFM and RFM-derived features, applies K-Means and Gaussian Mixture Models, and incorporates explainable AI techniques to interpret cluster formation. Additionally, comprehensive visualization strategies with cluster profile using radar charts and feature importance representations—are employed to enhance business interpretability. The optimal number of clusters is determined using Silhouette Score, Elbow Method, and information-theoretic criteria (AIC and BIC), ensuring both statistical robustness and practical relevance.

LITERATURE REVIEW

This section reviews the theoretical foundations and empirical studies that form the basis of this research. The discussion is structured into three key areas: the fundamental concepts of customer segmentation in the e-commerce context, the application of the RFM (Recency, Frequency, Monetary) model for behavioral feature extraction, and an overview of unsupervised machine learning algorithms, specifically Clustering, used to identify distinct customer groups.

Customer Segmentation

Customer segmentation is a process of dividing the customer group into subgroups according to their similarities (Rungruang et al., 2024; Choi et al., 2020). Basically, a simple customer segmentation approach includes geographic, demographic, psychographic, and behavioral segmentation (Rungruang et al., 2024). Customer segmentation divides customers into groups based on their similar needs, characteristics, or behaviors, in order to maintain customer relationships and increase profits. This segmentation is applied

because each consumer has their own preferences and requires a different approach (RevoUpedia, 2023b). It is this difference in approach that makes it impossible for companies to apply a single strategy to all types of segments (RevoUpedia, 2023b). Therefore, once companies understand the preferences and uniqueness of each customer, they can prioritize offers related to those needs to be included in their marketing strategy (Tabianan et al., 2022).

In the initial step, information regarding customer transactions is stored in the form of data that will later be utilized in the pre-processing stage (Nalla and Reddy, 2021). Naturally, some of the data that undergoes several processes will not be used because it is unsuitable or contains anomalies. Only after segmentation is visible can companies more easily identify customer needs, enabling the development of more targeted products or services. In addition, companies can also identify the likelihood of customers purchasing their products or services (MarkPlus Institute, 2024).

RFM Model

RFM (Recency, Frequency, Monetary) analysis is a powerful and recognized technique in database marketing (Christy et al., 2021). This approach essentially aims to rank customers based on their prior purchasing history. RFM analysis finds use in a wide range of applications involving a large number of customers such as online purchases and retailing. This approach was introduced in 1994 by Hughes Back (Exsight, 2024).

The first attribute is **Recency**, which is used to measure customer value based on their last purchase. Recency value relates to the time elapsed between the date of the last transaction and the analysis period. The closer the last transaction is, the more loyal the customer is considered to be. Therefore, the most important data needed to calculate recency value is the date of the last transaction.

The next attribute is **Frequency**, which focuses on assessing how many transactions customers make. The more transactions they make, the greater their potential influence on a company (Exsight, 2024). The last attribute is **Monetary**, defined as the total amount of transactions made by customers in a given period (Christy et al., 2021). The higher the amount of money spent, the more revenue is generated for the company. By using variables such as Recency (R), Frequency (F), and Monetary (M), this method is expected to analyze the entire customer population without the need to create a curated sample set. This provides a significant advantage because the analysis is performed on comprehensive data, allowing businesses to gain more accurate insights into customer purchasing behavior. In addition, the RFM model relies on efficient and accurate data, leaving no room for human error, making the analysis results more reliable (Exsight, 2024).

Clustering

Clustering is an unsupervised learning approach that focuses on assessing graphical representations and internal indices. Clustering identifies intrinsic patterns in data and groups them based on certain similarities (RevoUpedia, 2023a). One example is the silhouette score, an advanced graphical representation that provides visual information about the quality of the partition (Rungruang et al., 2024). Data clustering entails the grouping of data points within a dataset based on shared characteristics. Various algorithms exist for this purpose, each suited for specific conditions (Amutha and Khan, 2023).

Clustering can be implemented using several algorithms such as K-Means, GMM, and others. However, not all clustering algorithms are suitable for every type of data or every analysis objective, which necessitates selecting the algorithm that best suits the specific needs. Clustering is created with the aim of facilitating the identification of patterns, trends, and relationships between data (RevoUpedia, 2023a). In addition, clustering techniques can also be used to detect anomalies by identifying data points that do not fit the general data pattern. Therefore, after completing the clustering, it is necessary to validate the results to check whether the clusters produced are relevant to the analysis objectives (RevoUpedia, 2023a).

METHODS

This study adopts a quantitative research design utilizing unsupervised machine learning algorithms to uncover latent customer segments within e-commerce transactional data. The research methodology is structured systematically, encompassing data acquisition, rigorous preprocessing, feature engineering based on the RFM model, dimensionality reduction, and comparative clustering analysis (Amutha and Khan, 2023). The overall research framework proposed in this study is illustrated in **Figure 1**.

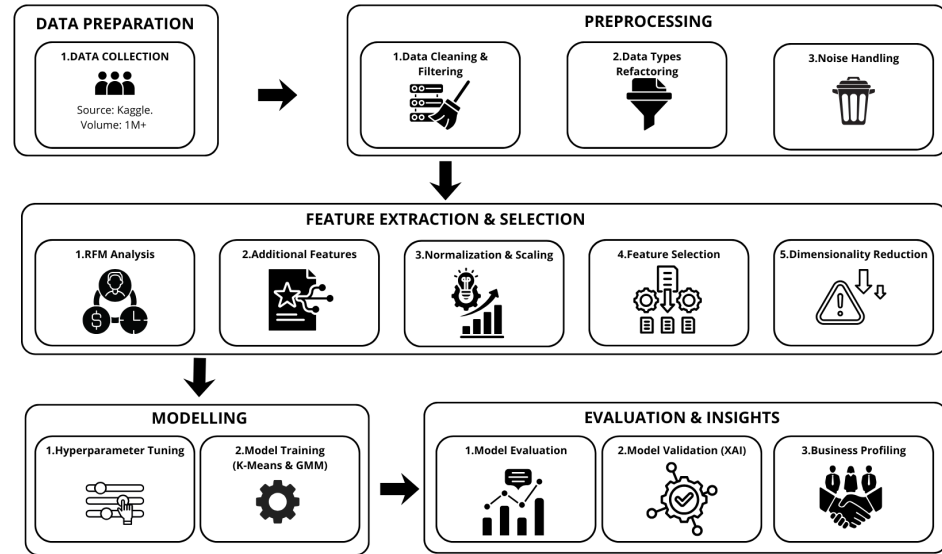


Figure 1. Proposed Research Pipeline

Data Collection

The empirical foundation of this study is the "Online Retail II" dataset, a comprehensive transactional archive retrieved from the public data repository, Kaggle (Hosseini and Shabani, 2022). This dataset constitutes a robust collection of historical purchase records generated by a UK-based and registered non-store online retail entity, primarily specializing in unique all-occasion gifts. The temporal scope of the data encompasses a two-year period, specifically spanning from December 1, 2009, to December 9, 2011.

In its raw, unprocessed state, the dataset comprises a substantial volume of 1,067,371 transactional rows. Each record represents a granular line-item transaction characterized by eight primary attributes: Invoice (transaction identifier), StockCode (product identifier), Description (product name), Quantity (order volume), InvoiceDate (timestamp), Price (unit cost), Customer ID (unique client identifier), and Country (customer location). This specific dataset was purposively selected due to its high degree of granularity and volume, which provides the statistical density required to model complex, non-contractual customer behavioral patterns effectively.

Data Preprocessing

Data preprocessing was conducted to ensure the reliability and analytical validity of the transactional data prior to customer behavior modeling (Nalla and Reddy, 2021). The process began with the identification of individual customers as a prerequisite for customer-centric segmentation. Consequently, records lacking valid CustomerID or Description values were removed, as such entries cannot be reliably associated with identifiable customers or meaningful product transactions. In addition, exact-match duplicate records were eliminated to prevent artificial inflation of transaction frequency in subsequent RFM-based feature construction.

Following the initial filtering, the dataset was further refined to retain only valid sales transactions. Records with Invoice identifiers prefixed by the letter 'C', which denote cancelled or returned orders, were excluded. Logical validation was also applied to quantitative attributes by removing transactions with non-positive values for Quantity or UnitPrice, ensuring financial consistency. Furthermore, non-merchandise stock codes associated with administrative or operational charges (e.g., 'POST', 'D', 'M', and 'BANK CHARGES') were filtered out to preserve the focus on genuine consumer purchasing behavior.

Finally, distributional anomalies were addressed to mitigate the influence of extreme values. The transactional data exhibited a heavy-tailed distribution, particularly in the Quantity feature, which can disproportionately affect distance-based clustering algorithms such as K-Means. To address this issue, Winsorization was applied to the upper tail of the Quantity distribution using a threshold of 0.1%.

183 This approach caps extreme values while retaining all observations, thereby reducing the impact of
184 bulk-purchase outliers without discarding potentially informative transactions or altering the underlying
185 distribution structure.

186 **Feature Extraction & Selection**

187 The transition from raw transactional records to actionable insights necessitates the aggregation of data to
188 a customer-centric level. This study implements a rigorous feature engineering pipeline to construct a
189 comprehensive behavioral profile for each of the 5,852 unique customers identified in the clean dataset.
190 The process encompasses feature extraction, statistical transformation, and dimensionality reduction.

191 **Feature Extraction**

192 Beyond the traditional RFM (Recency, Frequency, Monetary) framework, this research derives additional
193 behavioral metrics to capture the nuances of customer loyalty and spending consistency (Christy et al.,
194 2021). The feature extraction process yielded five quantitative dimensions. The core analysis rests
195 on the standard RFM metrics: Recency (R), which quantifies customer engagement by calculating the
196 days elapsed since the last valid transaction, where a lower value indicates higher recent engagement;
197 Frequency (F), representing the purchasing activity level defined by the count of unique invoices; and
198 Monetary (M), which measures the total revenue contribution calculated as the summation of the product
199 of quantity and unit price for all transactions ($M = \sum(Quantity \times Price)$) (Yan and Zhao, 2021).

200 To augment these traditional metrics, two derived features were introduced to provide deeper behav-
201 ioral insights. Tenure (T) was calculated to capture the longevity of the customer relationship, measured
202 as the duration in days between the customer's first recorded transaction and the dataset's snapshot date
203 (Pratiwi et al., 2025). Additionally, the Coefficient of Variation (CV) of Spending was computed to
204 assess the stability of a customer's purchasing power. Defined as the ratio of the standard deviation of
205 transaction values to the mean transaction value, this metric distinguishes between consistent spending
206 habits (indicated by a lower CV) and sporadic or variable purchasing behavior (indicated by a high CV).

207 **Statistical Transformation and Scaling**

208 Exploratory data analysis revealed that the distribution of the extracted features, particularly Monetary
209 and Frequency, followed a heavy-tailed (power-law) distribution with significant positive skewness. To
210 mitigate the adverse effects of skewness on distance-based clustering algorithms, a Logarithmic Trans-
211 formation ($\ln(x + 1)$) was applied to all features (Abbasimehr and Shabani, 2021). This transformation
212 effectively normalizes the data distribution and stabilizes variance.

213 Subsequently, to ensure that no single feature dominates the Euclidean distance calculations due
214 to magnitude differences (e.g., comparing Monetary in the thousands vs. CV Spending in decimals),
215 Min-Max Scaling was utilized. This process standardizes all features into a uniform range of [0, 1] using
216 the formula:

$$217 \quad X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (1)$$

218 **Feature Selection and Dimensionality Reduction**

219 To enhance model performance and reduce multicollinearity, a correlation analysis was conducted using a
220 Pearson Correlation Matrix. The analysis indicated a high redundancy between Frequency and Monetary.
221 Consequently, Frequency was excluded to prioritize financial value and spending behavior consistency.
222 The final feature set selected for modeling comprises four key dimensions: Recency, Monetary, Tenure,
223 and CV Spending.

224 Furthermore, to visualize the high-dimensional structure of the data and facilitate the identification
225 of natural clusters, this study employs Uniform Manifold Approximation and Projection (UMAP) as
226 implemented in recent customer segmentation studies (Bezerra and Silva, 2020). Unlike linear methods
227 such as PCA, UMAP preserves both local and global data structures. The algorithm was configured
228 with specific hyperparameters: $n_components = 2$ for 2D projection, $n_neighbors = 15$ to balance local
229 structure preservation, and $min_dist = 0.1$ to control the tightness of the clustered points.

230 **Modeling Strategy**

231 To identify distinct customer segments with high precision, this research employs a comparative analysis
232 of two prominent unsupervised machine learning algorithms: K-Means Clustering and Gaussian Mixture

Models (GMM) (John et al., 2023). While K-Means is widely recognized for its computational efficiency in partitioning data, GMM offers a probabilistic approach that accommodates clusters of varying geometric shapes. The modeling procedures are detailed as follows:

K-Means Clustering

This study employed K-Means Clustering, a centroid-based algorithm that partitions the dataset X into K disjoint clusters by minimizing the Within-Cluster Sum of Squares (WCSS), also referred to as inertia (Miraftabzadeh et al., 2023). The primary objective is to minimize the sum of squared Euclidean distances between each data point and its corresponding cluster centroid. To ensure model stability and reproducibility, the algorithm was rigorously configured with specific hyperparameters. The ‘k-means++’ method was utilized for centroid initialization to accelerate convergence and mitigate the risk of falling into suboptimal local minima (Sinaga and Yang, 2020). Furthermore, the model was executed with $n_{init} = 40$, meaning the algorithm ran 40 times with different centroid seeds to return the model with the lowest inertia, while the maximum number of iterations (max_iter) was set to 300 to guarantee convergence even for complex data distributions (Yunita and Ali, 2025).

The determination of the optimal number of clusters (K) was conducted using a dual-validation approach to ensure statistical robustness. First, the Elbow Method was employed to visualize the decay of WCSS as K increases, identifying the “elbow” point where the marginal gain in variance reduction diminishes significantly (Marisa et al., 2021). Complementing this, Silhouette Analysis was performed to calculate the mean Silhouette Coefficient for each K (Thakkar and Varma, 2021). This metric evaluates clustering quality based on cohesion (distance to own cluster) and separation (distance to nearest neighbor cluster), where a score closer to +1 indicates well-defined clusters.

Gaussian Mixture Model (GMM)

To address the limitations of K-Means specifically its assumption of spherical cluster shapes and equal variance this study subsequently applied the Gaussian Mixture Model (GMM) (Alves Gomes and Meisen, 2023). GMM is a probabilistic model assuming that all data points are generated from a mixture of a finite number of Gaussian distributions with unknown parameters. The model parameters were estimated using the Expectation-Maximization (EM) algorithm (Zhang et al., 2021). Regarding configuration, the covariance parameter was set to ‘full’ ($covariance_type = 'full'$) (Wang et al., 2024). This setting allows each component to possess its own general covariance matrix, enabling the model to capture clusters with elliptical shapes and varying orientations. This flexibility is crucial for RFM data, which often exhibits significant correlation between features, such as Monetary and Frequency. Additionally, to accelerate EM convergence, the initial weights, means, and precisions were initialized using the results from K-Means ($init_params = 'kmeans'$).

Unlike K-Means, model selection for GMM relies on information-theoretic criteria rather than inertia to balance model fit and complexity, thereby preventing overfitting. Two metrics were minimized to identify the optimal component count: the Akaike Information Criterion (AIC), which estimates the relative quality of the statistical models, and the Bayesian Information Criterion (BIC), which acts similarly to AIC but imposes a stronger penalty for models with a higher number of parameters. The optimal number of components is identified at the point where both AIC and BIC scores are minimized.

Model Evaluation and Explainability

To ensure both the statistical robustness and the interpretability of the derived customer segments, this research adopts a comprehensive evaluation framework. This framework consists of quantitative internal validation, model validation using Explainable AI, and qualitative cluster profiling.

Quantitative Evaluation Metrics

Since ground truth labels are unavailable in unsupervised learning, the quality of the clustering results is evaluated based on the intrinsic properties of the data, specifically cohesion (compactness within clusters) and separation (distinctness between clusters). To this end, three standard metrics were employed. First, the Silhouette Score measures how similar an object is to its own cluster (cohesion) compared to other clusters (separation) (Thakkar and Varma, 2021). The silhouette coefficient for a data point i is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (2)$$

282 Where $a(i)$ is the mean distance between i and all other points in the same cluster, and $b(i)$ is the
283 minimum mean distance of i to all points in any other cluster. The score ranges from -1 to $+1$, where a
284 value closer to $+1$ indicates a well-defined and distinct cluster configuration.

285 In addition to the Silhouette Score, two other indices were utilized. The Calinski-Harabasz Index, also
286 known as the Variance Ratio Criterion, evaluates the ratio of the sum of between-cluster dispersion to
287 within-cluster dispersion (Baati and Hamadou, 2022). For this metric, a higher score signifies that the
288 clusters are dense and well-separated. Finally, the Davies-Bouldin Index was employed to measure the
289 average ‘similarity’ between clusters, comparing the distance between clusters with their sizes. Unlike
290 the previous metrics, a lower Davies-Bouldin index indicates a model with superior separation between
291 clusters (Baati and Hamadou, 2022).

292 **Model Validation Using Explainable AI (XAI)**

293 To validate that the formed clusters are not merely statistical artifacts but represent distinct behavioral
294 patterns driven by significant features, a post-hoc validation approach using Explainable AI (XAI)
295 was implemented. A Random Forest Classifier was trained as a surrogate model to predict the cluster
296 labels generated by the primary clustering algorithm (K-Means/GMM). The model was configured with
297 $n_estimators = 200$ to ensure stability. The classification accuracy of this surrogate model serves as a
298 validation metric; high accuracy confirms that the clusters are distinguishable based on the input features
299 (RFM+Tenure).

300 Subsequently, SHAP (SHapley Additive exPlanations) values were calculated to validate the feature
301 contribution (Mokhtari and Higdon, 2021). By analyzing the global feature importance, this step verifies
302 whether the segmentation is driven by relevant business metrics (e.g., Monetary or Recency) rather than
303 noise or irrelevant variables. This ensures the “trustworthiness” of the model before business decisions
304 are made.

305 **Cluster Profiling**

306 Following the validation phase, the characteristics of each cluster are analyzed to derive actionable
307 business insights. This profiling process involves calculating the central tendencies (mean and median) of
308 the original RFM and Tenure values for each group. By comparing these centroids against the overall
309 population average, each cluster is assigned a descriptive business label (e.g., “Loyal Customers,” “At-
310 Risk,” or “New Customers”) (Isnain et al., 2025). This step bridges the gap between the mathematical
311 output of the algorithm and strategic marketing implementation.

312 **RESULTS**

313 This section presents the empirical results of customer segmentation using K-Means and Gaussian
314 Mixture Model (GMM) on e-commerce transaction data. The results include model evaluation, cluster
315 visualization, and segment profiling based on transaction behavior features.

316 **K-Means Clustering Results**

317 The implementation of the K-Means algorithm was conducted in two primary stages determining the
318 optimal number of clusters and building the final model. In the first experimental stage, which aimed to
319 determine the optimal K value, the model was configured with the parameters $n_init = 40$, $max_iter =$
320 300 , and $random_state = 0$ to ensure a balance between computational efficiency and the reliability of the
321 evaluation outcomes.

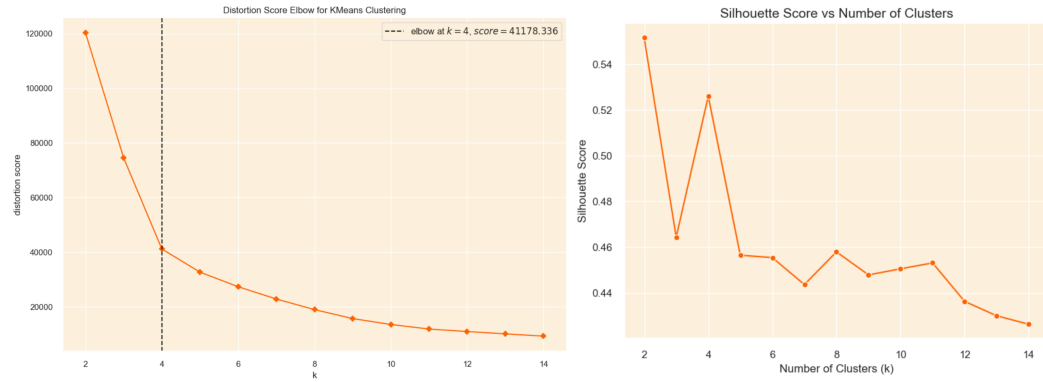


Figure 2. UMAP visualization of K-Means clustering results

Based on these experiments, the Elbow Method analysis in Figure 2 revealed a significant drop in distortion that started to stabilize at $k = 4$, with a distortion value of 41,178.336. This finding was validated by the Silhouette Score analysis, where the 4-cluster configuration yielded an average score of 0.526, indicating a reasonably solid clustering structure compared to other cluster counts.

Once the optimal number of clusters was determined $K = 4$, the final model was retrained using stricter parameters to ensure maximum convergence specifically with $n_{\text{init}} = 40$ and $\text{max_iter} = 300$ using `k-means++` initialization. The final evaluation demonstrated statistically valid results, with a Calinski-Harabasz Index of 10,825.30 indicating strong inter-cluster separation, and a Davies-Bouldin Index of 0.649 confirming good internal homogeneity within each formed customer segment.

To visually validate the clustering structure, this study employs the dimensionality reduction technique known as Uniform Manifold Approximation and Projection (UMAP). This technique projects complex customer behavior features into a two-dimensional space to facilitate the interpretation of inter-cluster separation.

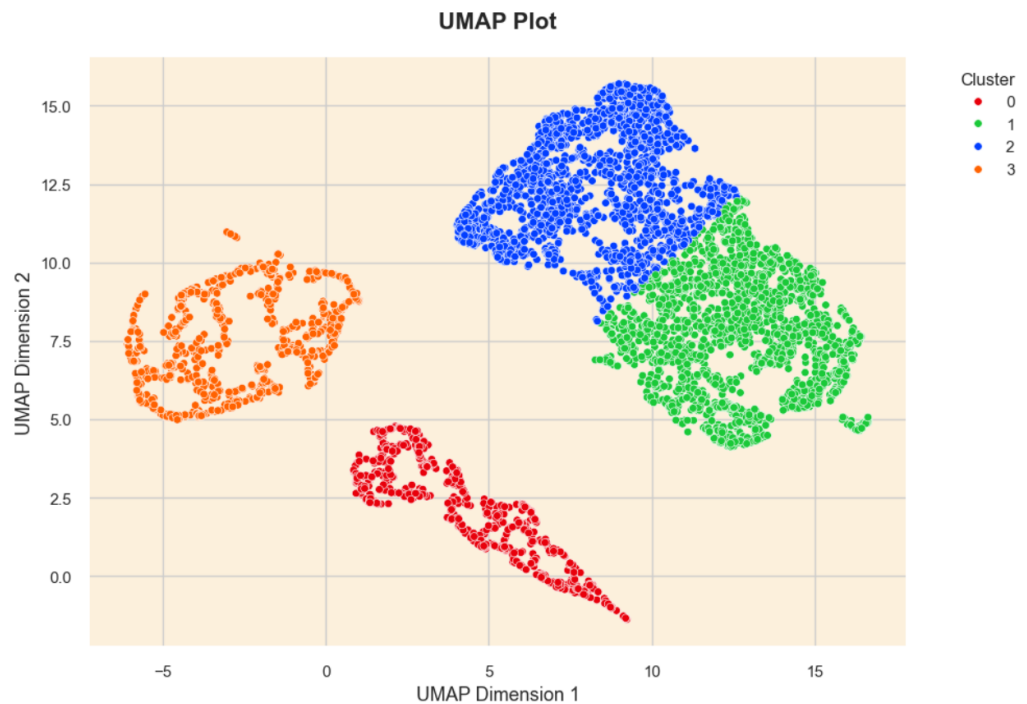


Figure 3. UMAP visualization of K-Means clustering results

As illustrated in Figure 3, the UMAP visualization reveals the formation of four distinct data groups that are clearly distributed and separated. There is no significant overlap between the clusters, with each color representing a unique customer segment. This visual evidence supports the findings from the previous evaluation metrics, confirming that the K-Means model successfully partitions customers into segments with distinct and distinguishable behavioral characteristics.

Gaussian Mixture Model Clustering Results

To serve as a comparison and validate the stability of the segments produced by K-Means, this study applied the Gaussian Mixture Model (GMM) algorithm. Unlike K-Means, which utilizes a hard clustering approach, GMM employs a probabilistic approach (soft clustering) in which each data point has a specific probability of belonging to each cluster.

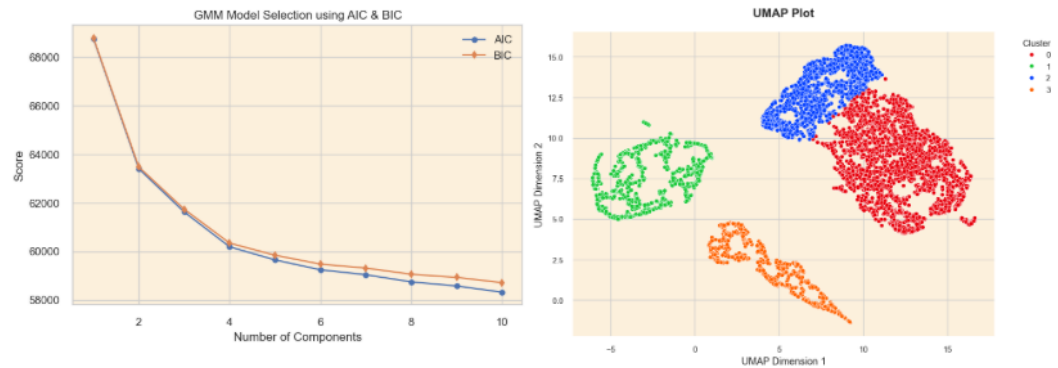


Figure 4. GMM Model Evaluation: Model Selection (AIC/BIC) and Cluster Visualization

The determination of the optimal number of components (clusters) in the GMM was based on information criteria, specifically the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). As shown in Figure 4, the plot of AIC and BIC values indicates a sharp downward trend up to the 4th component, after which it begins to plateau. Based on this pattern, and to maintain consistency for comparison with the K-Means model, the number of components was set to $K = 4$.

The GMM model was subsequently configured using a 'full' covariance matrix. This parameter choice enables the model to capture correlations between features within each cluster and form clusters with flexible geometric orientations (ellipsoidal), unlike the 'diag' type which restricts cluster shapes to be axis-aligned. Additionally, parameter initialization using the k-means method with $n_{\text{init}}=50$ was applied to accelerate convergence and ensure model stability.

The evaluation results show that the 4-component GMM performed well. It achieved a Silhouette Score of 0.509 and a Calinski-Harabasz Index of 9766.76 and Davies-Bouldin Index of 0.646. While K-Means obtained a slightly better Silhouette Score (0.526), the difference is very small.

Cluster Profiling Results

Based on the analysis using the Gaussian Mixture Model (GMM), this section describes the behavioral traits of each customer group. Profiling was conducted by analyzing the mean values (centroids) of the dominant features: Recency, Monetary, Tenure, and CV Spending. It should be noted that the resulting profiles closely match the findings from the K-Means method, confirming the reliability of the customer segmentation.

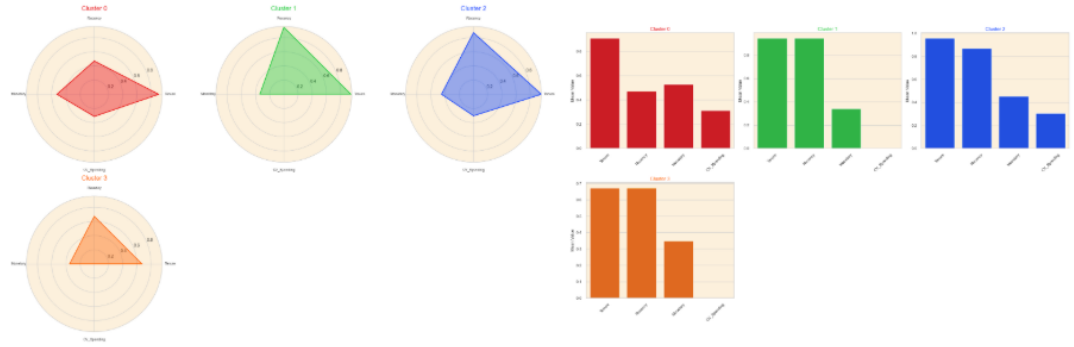


Figure 5. Customer Segment Profile Based on GMM Model

The visualization of these profiles through a Radar Chart in Figure 5 reveals four distinct customer segments. Cluster 0 (Loyal Customer) characterized by high Tenure and consistently high Monetary value, where a low Recency score indicates they remain highly active. In contrast, Cluster 1 (Lost Customer) consists of long-standing customers (high Tenure) who pose a high churn risk; this is evidenced by their very high Recency scores and low current Monetary contributions, indicating a long period of inactivity. Meanwhile, Cluster 2 (Dormant Loyal Customer) exhibits an intriguing pattern with high historical Tenure and Monetary values similar to loyal customers, yet currently showing high Recency, suggesting these high-value customers have paused their transactions. Finally, Cluster 3 (New Customer) comprises customers who have recently joined, characterized by low Tenure and low accumulated Monetary value due to limited transaction history, with Recency at a moderate level.

Explainable AI Results

To validate the integrity of the segments produced by the Gaussian Mixture Model (GMM) and to identify the most influential variables in cluster formation, this study applies an Explainable AI (XAI) approach. Since clustering algorithms lack native target variables, a surrogate model using a Random Forest Classifier was trained to predict the GMM cluster labels. Subsequently, SHapley Additive exPlanations (SHAP) values were calculated to measure the marginal contribution of each feature to the model's decisions.

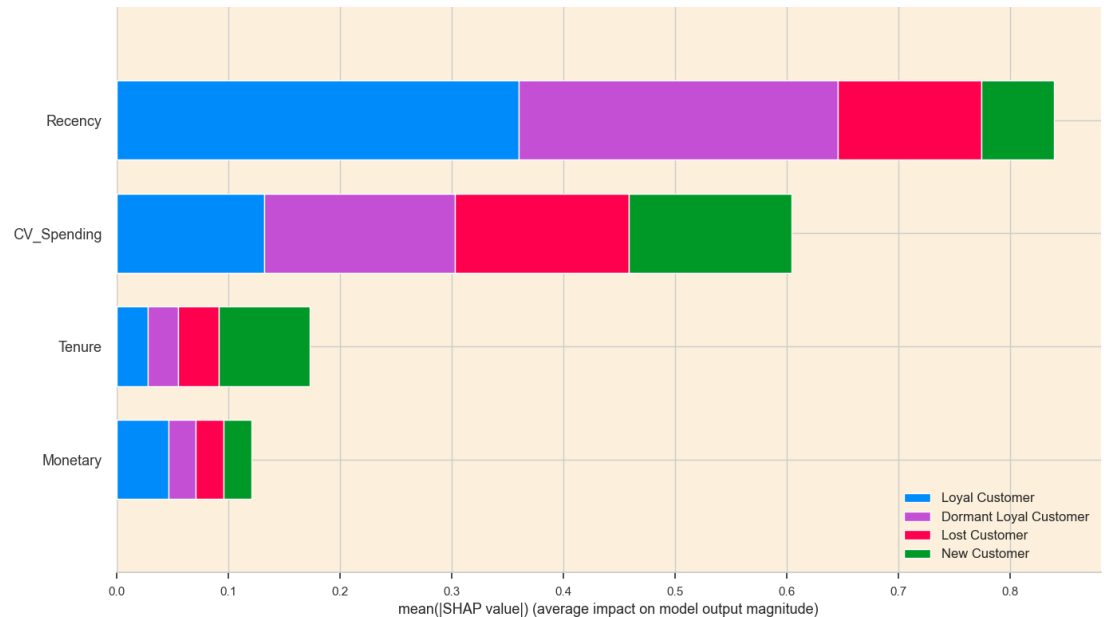


Figure 6. SHAP Summary Plot: Global Feature Importance

The analysis of feature importance, as illustrated in the SHAP Summary Bar Plot in Figure 6, establishes the hierarchy of dominant variables in distinguishing customer segments. Notably, Recency exerts the most significant influence on the clustering outcome, demonstrating the highest mean absolute SHAP value. This is followed in descending order of importance by CV Spending, Tenure, and Monetary.

From a behavioral perspective, this finding is highly intuitive in an e-commerce environment. Customers who have made purchases recently are naturally grouped into loyal or new customer segments, whereas those who have not transacted for long periods are driven toward dormant or lost customer segments. This result confirms that temporal purchasing behavior is more decisive than transactional value alone when distinguishing customer groups.

The second most important feature is CV Spending, which reflects the variability or consistency of a customer's spending behavior. Its high contribution suggests that how stable or volatile a customer's spending pattern is plays a major role in segmentation. Customers with high spending variability or dynamic purchasing behavior tend to be strongly associated with high-value segments such as Loyal and Dormant Loyal customers, whereas customers with low variability are more likely to belong to New or Lost customer segments. This implies that behavioral consistency is a stronger differentiator than sheer monetary volume.

Tenure ranks third in terms of influence. Although it is not as dominant as Recency or CV Spending, it remains highly informative for identifying newly acquired customers. Low Tenure values are strongly associated with clusters representing New Customers, confirming that account age is a structural feature that naturally separates recently onboarded users from long-term customers. This result indicates that the segmentation process successfully captures lifecycle stages of customers.

Interestingly, Monetary has the lowest overall impact among the four features. While spending amount is clearly important from a business perspective, this result indicates that financial value alone is not sufficient to explain behavioral segmentation. High Monetary values primarily support the classification of Loyal and Dormant Loyal customers, but they are not effective in distinguishing active versus inactive behavior. This reinforces the idea that Recency, rather than Monetary, is the decisive factor for determining customer activity status.

Overall, the SHAP results demonstrate that the dominance of Recency and CV Spending highlights the importance of transaction timing and spending consistency in shaping distinct customer segments. These findings validate that the proposed segmentation framework captures meaningful behavioral patterns, making it highly suitable for precision marketing strategies such as retention, reactivation, and onboarding campaigns.

DISCUSSION

The comprehensive customer segmentation analysis utilizing the Gaussian Mixture Model (GMM) on e-commerce transaction data offers profound insights into customer behavioral heterogeneity. This discussion focuses on interpreting the key findings derived from both the model evaluation and XAI validation, alongside formulating practical implications for business decision making.

Model Selection and Clustering Behavior

The adoption of GMM as the definitive model, notwithstanding the marginally superior Silhouette Score achieved by K-Means (0.526 vs 0.509), was predicated on its superior compatibility with the data's inherent characteristics. This choice was based on GMM's better fit with the data's natural features. Customer behavior segmentation often involves unclear and overlapping boundaries, such as the difference between Loyal and Dormant Loyal Customers. In this situation, GMM's probabilistic, or soft clustering, which supports a full covariance structure, is adjudged to be more realistic in representing this complexity than the absolute inherent in K-Means.

This decision was further supported by validation using Explainable AI (XAI) through SHAP, which identified Recency as the most important factor in distinguishing between customer segments. This has two main effects which is high Recency, meaning the customer has been inactive for a long time, strongly leads to classification as a Lost Customer, while low Recency results in being categorized as a Loyal Customer. The focus on Recency underscores that transactional recency is the most critical metric for assessing customer health within the e-commerce landscape.

Explainability and Feature Contribution

The SHAP values analysis reveals the crucial feature hierarchy for understanding customer behavior. Recency emerges as the most dominant predictor, indicating that, within this model, the timing of the last interaction is the strongest signal for determining customer status. Theoretically, this confirms that transactional 'freshness' carries more weight than historical past customers with high monetary value but poor Recency will be deemed high-risk.

Furthermore, CV Spending holds a unique role in measuring behavioral stability. This feature is vital because it differentiates between customers who shop regularly with stable values versus those whose spending is highly fluctuating.

Meanwhile, Tenure and Monetary function as intrinsic value determinants. Tenure provides a temporal context—distinguishing whether a 'low monetary value' is due to the customer being genuinely low-value or simply a recent entrant (New Customer). Monetary, conversely, remains the standard measure for customer asset valuation. The combination of these four features demonstrates that the model does not solely consider 'how much money' (Monetary), but prioritizes 'how active' (Recency) and 'how consistent' (CV Spending) a customer is.

Comparison with Previous Studies

The validity of the segmentation results in this study is reinforced through a visual comparison between our UMAP projection and conventional 3D Scatter Plots typically utilized in prior studies.

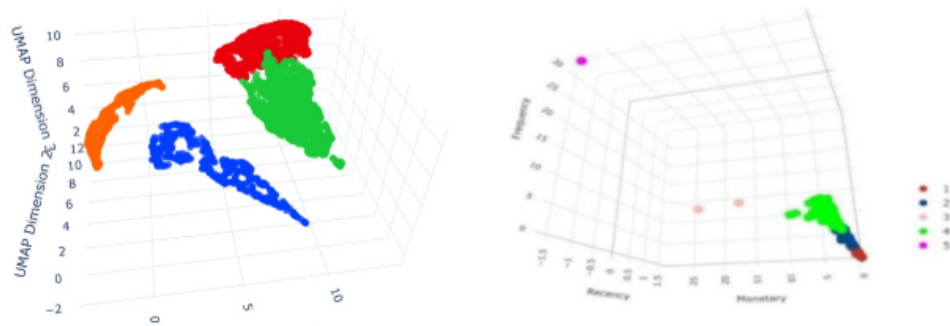


Figure 7. Visual Comparison of Cluster Structure

In comparative visualizations from other reports, the customer data distribution tends to aggregate densely in one corner with very vague boundaries between segments, making it difficult to differentiate the characteristics of most customers except for a few distant extreme outliers.

Conversely, our UMAP visualization shows a much clearer spatial separation, where the four clusters are perfectly resolved into isolated groups without significant overlapping. This contrasting difference indicates that can capture the complex structure of the data nuances and separate customer profiles more precisely than standard methods that only map raw data onto three dimensions, where the subtleties of intermediate segments are often hidden by data density.

Managerial Implications

The clear separation of customers into four different groups allows the company to transition from ineffective mass marketing strategies to a hyper-personalized approach. To ensure stability, Loyal Customers (Cluster 0) must remain the primary recipients of appreciation programs and exclusive rewards, with the strategic goal of maintaining a near-zero churn rate given their role as the revenue foundation. Concurrently, the Dormant Loyal (Cluster 2) segment represents high-value customers at risk, therefore, significant marketing resources should be allocated toward targeted win-back campaigns to reduce their high Recency status through bespoke incentives.

Furthermore, the strategy for New Customers (Cluster 3) requires an incubation approach centered on strong onboarding and product education rather than aggressive discounts, aiming to accelerate their growth in Tenure and Monetary metrics. Finally, identifying Lost Customers (Cluster 1) enables the

company to exercise budget efficiency by reducing wasted promotional spending on this low-potential group. These recouped resources can then be strategically reallocated toward the higher-potential Dormant Loyal or New Customer segments.

Limitations and Future Research

This study establishes a robust groundwork for subsequent research endeavors. First, future studies could focus on Dynamic Prediction by developing a supervised learning model to predict churn probability specifically for the Dormant Loyal segment, enabling real-time intervention strategies. Furthermore, External Data Integration offers a pathway to enhance current features, allowing for a better understanding of the factors driving behavior changes between clusters and leading to more accurate insights. Finally, conducting Product Analysis through product-level segmentation would help identify specific purchasing behaviors, such as cross-selling and up-selling opportunities relevant to each cluster, thereby resulting in more effective product recommendations.

CONCLUSIONS

This study demonstrates the effectiveness of an unsupervised machine learning framework for customer segmentation using large-scale e-commerce transaction data. By integrating RFM-based behavioral modeling, derived customer features, and probabilistic clustering, the proposed approach successfully identifies meaningful customer segments that reflect distinct stages of the customer lifecycle.

A comparative evaluation between K-Means and Gaussian Mixture Models (GMM) indicates that both algorithms produce robust and consistent segmentation structures. Although K-Means achieved slightly higher numerical validation scores, GMM was selected as the final model due to its ability to model complex data distributions through soft clustering, which is more representative of real-world customer behavior. The consistent identification of four optimal clusters across multiple validation methods further confirms the stability of the segmentation.

The four customer segments identified are Loyal, Dormant Loyal, Lost, and New Customers, which provide actionable insights for targeted marketing strategies, including retention, reactivation, and onboarding initiatives. Importantly, the incorporation of Explainable AI (XAI) through SHAP enhances the transparency of the clustering process by revealing that Recency and spending consistency are the dominant factors driving segment differentiation. This interpretability bridges the gap between advanced machine learning models and practical business decision-making.

Overall, this research contributes to the customer segmentation literature by extending traditional RFM analysis with derived behavioral features, probabilistic clustering, and explainable modeling. The proposed framework offers a scalable and interpretable solution for data-driven personalization strategies in modern e-commerce environments.

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